Shiyuan Huang

Ph.D. Student · Computer Science and Engineering

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Education

University of California, Santa Cruz — *Santa Cruz, California*

Ph.D. in Computer Science and Engineering — *2024 – Present*

- Advisors: Dr. Leilani H. Gilpin and Dr. Ian Lane
- Research Interests: Explainability and Trustworthiness in Language Models

University of California, Santa Cruz — *Santa Cruz, California*

M.S. in Computer Science and Engineering — *2022 – 2024*

- Advisor: Dr. Leilani H. Gilpin
- Graduated with a GPA of 3.97/4.00

University of California, Santa Cruz — *Santa Cruz, California*

B.S. in Computer Engineering — *2018 – 2022*

- Graduated with Highest Honor (Summa Cum Laude)
- Graduated with a major GPA in major coursework of 3.92/4.00

Professional Experience

2026.06-09	**Applied Scientist Intern**, Customer Service Data Intelligence, Amazon, Seattle
2025-	**Graduate Student Researcher**, Baskin School of Engineering, UC Santa Cruz
2022-	**Graduate Teaching Assistant**, Baskin School of Engineering, UC Santa Cruz
2021-2022	**Undergraduate Tutor, Reader**, Baskin School of Engineering, UC Santa Cruz
2021	**Software Engineer Intern**, Maxen Technology, Montréal, Québec, Canada

Publications

Shiyuan Huang, Li Liu, Jincheng He, Leilani H. Gilpin. 2026. Learning to Draw ASCII Improves Spatial Reasoning in Language Models. Conference on Language Modeling (COLM 2026)

Sicong Huang, Jincheng He, **Shiyuan Huang**, Karthik Raja Anandan, Arkajyoti Chakraborty, and Ian Lane. 2025. UCSC at SemEval-2025 task 3: Context, models and prompt optimization for automated hallucination detection in LLM output. In Proceedings of the 19th International Workshop on Semantic Evaluation (SemEval-2025).

Shiyuan Huang, Siddarth Mamidanna, Shreedhar Jangam, Yilun Zhou, Leilani H. Gilpin. 2023. Can Large Language Models Explain Themselves? A Study of LLM-Generated Self-Explanations. arXiv:2310.11207.

Research Experience

Branch to Learn: Improving Small Language Model Agents Without Task Rewards — *Seattle, WA; Santa Cruz, CA*

Amazon (Applied Scientist Intern) and UC Santa Cruz — *Jun. 2026 -*

- Calling a large model at every step of an agent workflow is costly at industrial data volumes, and the results of many workflows are hard to verify. Trained a small-language-model agent from the scores of a larger judge model alone, with no task rewards.
- The judge need not be a detector: branching at the agent's least-certain step trains agents as strong as branching at the step the judge names as faulty, so the judge only has to grade finished trajectories.
- Built a loop that keeps improving the agent on the tasks it serves, producing an SLM-based agent that narrows the gap to larger models at a fraction of their cost. Manuscript in preparation for the NAACL 2027 Industry Track.

- Began during my Applied Scientist internship at Amazon (Jun. 2026 - Sept. 2026); continuing at UC Santa Cruz.

Learning to Reason with Verifiable Intermediate Representations *Santa Cruz, CA | Jul. 2025 -*

- **Beyond spatial reasoning** (Sept. 2026 -). Extending checkable intermediate representations to other problems, so that each reasoning step can be verified by a program and used as a training signal.
- **Learning to construct layouts improves spatial reasoning (COLM 2026; Jul. 2025 - 2026).** Built Text2Space, 20,000 text descriptions paired with ground-truth ASCII layouts, which separates failures in constructing a spatial representation from failures in reasoning over it. All six LLMs tested read layouts 13 to 27 points better than they construct them. Training models to construct layouts improved spatial reasoning from text alone on all three external benchmarks (up to 43 points), with no layout at inference.

Evaluating the Reasoning and Explanations of Language Models *Santa Cruz, CA | Jun. 2023 -*

- **Benchmarking reasoning quality** (Apr. 2026 -). Developing a benchmark that scores the quality of a model's reasoning itself, including reasoning that is persuasive yet leads to a wrong conclusion.
- **Knowledge boundaries** (Sept. 2024 -). Quantifying language model familiarity across tasks and domains with likelihood analysis; studying abstention and linguistic biases.
- **LLM self-explanations** (2023 - 2024; arXiv:2310.11207). To our knowledge, the first rigorous study of LLM-generated feature-attribution explanations, compared with occlusion and LIME on faithfulness and agreement metrics.

Hallucination Detection in LLM Output (SemEval-2025 Task 3) *Santa Cruz, CA | Jan. 2025 - Apr. 2025*

- Implemented the system-combination component, which treats five systems as annotators and derives a confidence score from their agreement.
- Best average rank across 14 languages among 43 teams; Best System Paper Award for Task 3.

Presentations

Contributed Presentations

Shiyuan Huang, Li Liu, Jincheng He, Leilani H. Gilpin. 2026. Learning to Draw ASCII Improves Spatial Reasoning in Language Models. COLM 2026: San Francisco, California, October 2026.

Shiyuan Huang. Self-Explanations in Large Language Models. NVIDIA GTC 2026.

Sicong Huang, Jincheng He, **Shiyuan Huang**, Karthik Raja Anandan, Arkajyoti Chakraborty, and Ian Lane. 2025. Context, Models and Prompt Optimization for Automated Hallucination Detection in LLM Output. BayLearn 2025: Santa Clara, California.

Sicong Huang, Jincheng He, **Shiyuan Huang**, Karthik Raja Anandan, Arkajyoti Chakraborty, and Ian Lane. 2025. UCSC at SemEval-2025 task 3: Context, models and prompt optimization for automated hallucination detection in LLM output. The 19th International Workshop on Semantic Evaluation, ACL 2025: Vienna, Austria.

Shiyuan Huang, Siddarth Mamidanna, Shreedhar Jangam, Yilun Zhou, Leilani H. Gilpin. 2024. Can Large Language Models Explain Themselves? A Study of LLM-Generated Self-Explanations. BayLearn 2024: Apple, Cupertino, California.

Shiyuan Huang. 2024. Exploring Explainability and Interpretability in Generative AI. YRRSDS 2024; SIGDIAL 2024: Kyoto University, Kyoto, Japan.

Awards, Fellowships, & Grants

2025 **Best System Paper Award for Task 3, SemEval-2025**, The 19th International Workshop on Semantic Evaluation, Vienna, Austria

2024 **Deans' and Chancellor's Research Award**, Spring Quarter, UC Santa Cruz

2022 **Dean's Honor**, Spring Quarter, UC Santa Cruz

2021 **Dean's Honor**, Fall Quarter, UC Santa Cruz
Dean's Honor, Spring Quarter, UC Santa Cruz

2020 **Dean's Honor**, Fall Quarter, UC Santa Cruz
Dean's Honor, Summer Quarter, UC Santa Cruz
Dean's Honor, Winter Quarter, UC Santa Cruz

2019 **College Scholar Program**, UC Santa Cruz
Dean's Honor, Fall Quarter, UC Santa Cruz

Projects

AI Purchase Copilot (Analyzer) *Santa Cruz, CA*
Advisor: Dr. Ian Lane *Jul. 2025 - Mar. 2026*
- Designed and developed, with UCSC Financial Affairs, an LLM copilot that helps procurement professionals identify purchase requests that need further review, using knowledge injection, feedback learning, RAG, and fine-tuning.

BELS Assets Management System *UC Santa Cruz*
Advisor: Dr. Richard K Jullig *Jan. 2022 - Jun.2022*
- Developed a web-based system for Baskin Engineering Lab Support (BELS) that serves as a centralized platform for storing detailed information about assets and monitoring their current status

Baskin School of Engineering Course Scheduling Application *UC Santa Cruz*
Advisor: Dr. Richard K Jullig *Sept. 2021 - Dec.2021*
- Designed an application that monitors students' academic progress and offers personalized recommendations to enhance productivity

Maxen Technology *Montréal, Québec, Canada*
Software Development Intern *Jun. 2021 - Sept.2021*
- Collaborated with the software development team to design and implement backend endpoints using Django.

Teaching Experience

2026 **Beginning Programming in Python**, Teaching Assistant
2024 **Programming Abstractions: Python**, Teaching Assistant
2023 **Programming Abstractions: Python**, Teaching Assistant
Fall 2022 **Programming Abstractions: Python**, Teaching Assistant
2022 **Beginning Programming in Python**, Tutor, Reader
2021 **Beginning Programming in Python**, Tutor, Reader

Outreach & Professional Development

Peer Review

Reviewer, BayLearn Symposium 2025 (8 papers) and 2026 (5 papers)

Professional Memberships

Student Member, American Society for Engineering Education (ASEE)
Student Member, Association for Computing Machinery (ACM)